

MACHINE LEARNING -

Assignment -1

Polynomial Regression:

Finding the right polynomial for two datasets

GithubRepo: https://github.com/tomstanly1234/BT2024080_ML

Modules used: Numpy, Pandas, scikit learn

Files Submitted:

Training1.py; Training2.py, BT2024080_test_var2.csv,BT2024080_train_var2.csv,
BT2024080_test_var1.csv, BT2024080_train_var1.csv

Problem Statements:

- 1) **Power Plant Steam Turbine Optimization (var1) Scenario:** Before drilling new production wells, you must optimize the surface energy generation facility. Your plant operates a multi-stage steam turbine system. Due to slight variations in manufacturing tolerances and local water chemistry across regional plants, your specific plant setup behaves uniquely compared to your colleagues' facilities. The turbine efficiency is governed by six key operational parameters (x1, x2, x3, x4, x5, x6), representing percentage deviations:

x1 High-pressure steam valve adjustment

x2 Condenser coolant flow rate adjustment

x3 Re-injection pump hydraulic pressure

x4 Turbine blade pitch angle

x5 Non-condensable gas exhaust valve rate

x6 Steam inlet pressure adjustment

2) Phase 2: Subterranean Thermal Reservoir Mapping (var2) Scenario: With your surface plant optimized, Phase 2 requires you to select the exact coordinates for drilling new geothermal extraction wells. You are assigned to map a specific, high-potential geological survey block. You deploy seismic and thermal probe sensors across a 3D grid around your central basecamp. The variables x1, x2, and x3 represent spatial coordinate offsets in meters from the site landmark:

- x1: East-West coordinate offset**
- x2: North-South coordinate offset**
- x3: Vertical depth offset relative to the basecamp**

At each 3D coordinate, the sensor network records a Thermal Anomaly Score (y). Because of complex geological structures, the heat map in your sector resembles a highdegree polynomial (up to degree 20). Your Task: Due to survey budget limits, you have only collected physical core samples at a random scatter of 3D points (train). You must build a model to predict the Thermal Anomaly Score across the remaining planned drilling sites (test) to identify the exact coordinates and depth for drilling high-yield wells.

Submitted by:

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Polynomial Regression Models

Overview

This project develops regression models for predicting the target variable y from the given input features.

Two separate problems are considered:

- **VAR1:** Uses $x_1, x_2, x_3, x_4, x_5, x_6$ to predict y
- **VAR2:** Uses x_1, x_2, x_3 to predict y

The models use polynomial feature expansion, regularization, and cross-validation to select an appropriate model complexity while avoiding unnecessary overfitting.

Project Structure

```
.| files.csv
|  └─ train_var1.py
|  └─ train_var2.py
└─ README.md
```

Methodology

Problem 1 — VAR1

The first problem uses six input variables:

$x_1, x_2, x_3, x_4, x_5, x_6$

to predict:

y

Polynomial features are generated for different degrees. Since polynomial expansion can produce a large number of features, **Lasso regression** is used as a regularization technique.

The Lasso objective is:

Which is essentially the Loss Function + a constant $\lambda * (|w_1| + \dots + |w_7|)$

Lasso helps control model complexity and can reduce the influence of unnecessary polynomial terms by shrinking coefficients toward zero.

Degree: 5

Alpha : 0.01

CV MSE: 0.35538397464664834

CV R2 : 0.962428462568086

Problem 2 — VAR2

The second problem uses:

x_1, x_2, x_3

to predict:

y

A **Chebyshev polynomial basis** is used instead of ordinary polynomial features. The input variables are first scaled approximately to the range $[-1, 1]$. The reason I used this is because I faced issues with very large degree polynomials and their interactions. This makes sure that our information is scaled to a smaller value and also the relationship between values is made in a much more easier way using certain bases.

This allows both individual nonlinear effects and interactions between the input variables.

Ridge regression is used to regularize the resulting polynomial model:

The number of variables are low, so rather than eliminating them we instead use ridge to make sure that the total norm of the coefficients is minimised since this complex dataset will require every interaction to be there.

Degree: 8

Alpha : 0.01

CV MSE: 0.2875586680200255

CV R2 : 0.9937517269174747

Model Selection

Cross-validation is used to select the polynomial degree and regularization parameter.

The training data is divided into **3 folds**.

For each candidate model:

1. Train on two folds.
2. Validate on the remaining fold.

3. Calculate validation MSE.
4. Calculate validation R^2 .
5. Repeat for all three folds.
6. Average the validation results.

The main metrics are:

Mean Squared Error

MSE

Lower MSE indicates better prediction accuracy.

R^2

Higher R^2 indicates that the model explains more of the variation in the target.

Choosing the Polynomial Degree

The model with the absolute lowest cross-validation MSE is not automatically selected.

A tolerance-based approach is used to avoid unnecessary model complexity.

Models whose MSE is within approximately **2% of the best CV MSE** are considered comparable. Among these models, the model with the **smaller polynomial degree** is preferred. I have done it so we can use the minimal degree without compromising much on the error we encounter.

For example, if:

Degree 4 → MSE = 0.320

Degree 6 → MSE = 0.315

the improvement may not justify the additional complexity, so the lower-degree model can be selected.

This provides a balance between:

- prediction accuracy
 - model complexity
 - computational cost
 - risk of overfitting
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Final Training

After selecting the degree and regularization parameter using cross-validation, the selected model is retrained using **all available training data**.

The final model is then applied to the test features to generate predictions for y .

The test data is not used during model selection.

Prediction Files

The final prediction files follow the required submission naming convention:

<ROLLNO>_pred_var1.csv

<ROLLNO>_pred_var2.csv

MINE:

BT2024080_pred_var1.csv

BT2024080_pred_var2.csv

The exact column format follows the sample submission provided for the assignment.

Running the Code

1. Install dependencies

```
pip install numpy,sklearn
```

2. Run VAR1

```
python training1.py
```

3. Run VAR2

```
python training2.py
```

The scripts perform model selection, train the final model, and generate the corresponding prediction CSV.

Requirements

The project uses Python and the following main libraries:

numpy

pandas

scikit-learn

So, A requirements.txt file can contain these modules before you run the code.

Reproducibility

A fixed random seed is used for cross-validation so that the data splits remain consistent between runs.

The complete preprocessing, feature construction, model selection, training, and inference procedure is implemented in the repository.

Important Notes

- Cross-validation MSE and R^2 are calculated using validation portions of the **training data**.
 - The official test set is not used for model selection.
 - If the test set does not contain the true y values, test MSE and R^2 cannot be calculated locally.
 - The final prediction files contain the model's predicted y values.
 - I have used other datasets as an additional cross-person generalization check, but it is not used to select the official model.
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